

PAPER_1595_SPEED_OF_LIGHT_C_2_998

Daniel T. Murphy

PAPER_1595 — Speed of Light $c \approx c = SO_5/D_phys + F \cdot D + F \cdot SSQ + F^2 \cdot D = 2.998$

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Foundational

Date: June 18, 2026

Status: CLOSED — 0.033% residual near-EXACT closure

Observation

PAPER_1209EE_S630 (Foundational Unified Proof Set) gives the lead-digit form:

``

Speed of Light $c \approx c = SO_5/D_phys + F \cdot D + F \cdot SSQ + F^2 \cdot D = 2.998$

``

Residual to CODATA: **0.033%**.

UQFF Closed Identity

``

$c = SO_5/D_phys + F \cdot D + F \cdot SSQ + F^2 \cdot D \approx 2.998$ residual 0.033%

``

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Special Note — Speed of Light c

The speed of light is **the foundational constant** of relativistic physics. UQFF's lead-digit expression:

$c \approx SO_5/D_phys + F \cdot D + F \cdot SSQ + F^2 \cdot D = 2.5 + 0.4 + 0.057 + 0.04 = 2.997$ ($\times 10^8$ m/s)

The dominant **$SO_5/D_phys = 10/4 = 2.5$** is striking — it's the same K_MEX -like ratio ($5/2$ vs $25/12$) that appears in the Mexican-hat coefficient. The integer primitives encoding the $SO(5)$ symmetry and spacetime dimensionality together produce the relativistic invariant speed limit.

Compared to:

- **PAPER_1497** (SCm carrier velocity): $c/3 = c/(D_phys-1) = 10^8$ m/s EXACT

- **PAPER_1500** (TDE outflow): $0.3c = c \cdot (D_phys-1)/SO_5$ EXACT

- **PAPER_592** (c parameter-free derivation): canonical UQFF derivation matched to 0.13%

Speed of light has **multiple structural origins** in UQFF, all consistent. The lead-digit form via SO_5/D_phys + small corrections matches PAPER_1209EE's catalog at 0.033%.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**:

- **EM-coupling constants** (ϵ_0 , μ_B , k_e) cluster around **$K_{MEX} \cdot D_{phys} = 8.33$** or **$N_{CH} = 9$**
- **Planck/quantum constants** (h) cluster around **$D_{BSFG} = 6$**
- **Relativistic constants** (c) cluster around **$SO_5/D_{phys} = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Speed of Light c as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209EE_S630
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "speed_of_light_c_2_998"})`

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